Grade 7 Creative Arts and Design

Strand 1: Design

Lesson 1 — Design in Nature and the Manmade Environment

Codes: B7.1.1.1.1 – B7.1.1.1.3 **Format:** Theory Questions and Answers

Preamble (pre-lesson)

Design is how things are planned so they work well and look good. You can see design in **nature** (leaves, birds, rivers) and in the **manmade world** (buildings, roads, cloth).

This lesson answers questions about:

1. What design means and why it matters.
2. The **elements** of design: dots, lines, shapes.
3. The **principles** of design: balance, rhythm, repetition.

Words to know first:

- **Design** — planning how something looks and works.
- **Nature** — the natural world of plants, animals, rivers, and rocks.
- **Manmade** — things made by people.
- **Element** — a basic building block of design.
- **Principle** — a rule for arranging elements.
- **Biomimicry** — copying ideas from nature to solve human problems.
- **Pattern** — a repeated design.

A. Design and why it matters

Q1. What is design? **Ans.** Design is planning how something looks and works.

Q2. What is nature? **Ans.** Nature is the natural world of plants, animals, rivers, and rocks.

Q3. What does "manmade" mean? **Ans.** Manmade means anything made by people rather than by nature.

Q4. Mention three reasons design is important. **Ans.**

- It solves problems.
- It communicates ideas.
- It expresses feelings and culture.

Q5. What is biomimicry? **Ans.** Biomimicry is copying ideas from nature to solve human problems.

Q6. Give one example of biomimicry. **Ans.** The beak of the kingfisher bird inspired the nose of the Shinkansen bullet train.

Q7. What manmade design did rivers inspire? **Ans.** Rivers inspired the design of winding roads.

Q8. What manmade design did anthills inspire? **Ans.** Anthills inspired house designs with natural air flow.

Q9. What natural design is similar to a fishing net? **Ans.** A spider web is a natural design similar to a fishing net.

Q10. How does design help learning? **Ans.** Design helps learning through clear posters, diagrams, and models.

B. Elements of design

Q11. What are the elements of design? **Ans.** The elements of design are dots, lines, and shapes.

Q12. What is a dot? **Ans.** A dot is a single small point.

Q13. What is a line? **Ans.** A line is a dot that moves in a direction.

Q14. What is a shape? **Ans.** A shape is a closed outline you can see.

Q15. Give two examples of a dot in nature. **Ans.**

- Seeds
- Stars in the sky

Q16. Give two examples of a line in nature. **Ans.**

- Plant stems
- The edge of a leaf

Q17. Give two examples of a shape in nature. **Ans.**

- Petals
- Stones

Q18. What is a geometric shape? **Ans.** A geometric shape is an exact shape such as a circle, square, or triangle.

Q19. What is an organic shape? **Ans.** An organic shape is a free-flowing natural shape such as a cloud or a leaf.

Q20. Mention the three elements of design and give one example of each. **Ans.**

- Dot — a seed
- Line — a plant stem
- Shape — a petal

C. Principles of design

Q21. What are three principles of design? **Ans.** Three principles of design are balance, rhythm, and repetition.

Q22. What is balance in design? **Ans.** Balance is when the parts of a design feel evenly arranged.

Q23. Give one example of balance in nature. **Ans.** The two sides of a face are an example of balance in nature.

Q24. What is rhythm in design? **Ans.** Rhythm is a sense of movement created by repeating elements.

Q25. Give one example of rhythm in manmade design. **Ans.** A row of windows on a building is an example of rhythm.

Q26. What is repetition in design? **Ans.** Repetition is using the same element again and again.

Q27. Give one example of repetition in cloth design. **Ans.** A repeated motif printed all over a cloth is an example of repetition.

Q28. How does repeating dots create rhythm? **Ans.** Repeating dots in a line makes the eye move along the line.

Q29. How do shapes create balance? **Ans.** Shapes create balance when they are arranged evenly on both sides.

Q30. What is a pattern? **Ans.** A pattern is a design that repeats in a regular way.

D. Nature and manmade design together

Q31. How is a river similar to a winding road? **Ans.** Both follow the shape of the land as they flow or pass through it.

Q32. How is a spider web similar to a fishing net? **Ans.** Both use a network of lines to catch things.

Q33. How are leaf veins similar to a road network? **Ans.** Both branch out from a main line to reach many points.

Q34. Mention two natural designs and two manmade designs. **Ans.**

- Natural designs: spider web, seashell
- Manmade designs: bridge, market stall

Text diagram — elements of design

```
DOT LINE SHAPE
. _____ / \
. / | |
. / \___/
points a moving dot closed outline
```

Text diagram — principles of design

```
BALANCE RHYTHM REPETITION
<--- equal ---> wave wave wave || || || ||
left right movement same thing again
```

Text diagram — biomimicry examples

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NATURE --> MANMADE
kingfisher beak --> bullet train nose
river path --> winding road
anthill air flow --> house ventilation
leaf veins --> road network
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Exercises (8 sections × 5 questions)

Exercise 1 — Meaning of design

1. What is design?
2. What is nature?
3. What does "manmade" mean?
4. Mention two reasons design is important.
5. **[Review]** What is biomimicry?

Answers

1. Design is planning how something looks and works.
2. Nature is the natural world of plants, animals, rivers, and rocks.
3. Manmade means anything made by people rather than by nature.
4. It solves problems; it communicates ideas.
5. Biomimicry is copying ideas from nature to solve human problems.

Exercise 2 — Elements of design

1. What are the elements of design?
2. What is a dot?
3. What is a line?
4. What is a shape?
5. **[Review]** What is a pattern?

Answers

1. The elements of design are dots, lines, and shapes.
2. A dot is a single small point.
3. A line is a dot that moves in a direction.
4. A shape is a closed outline you can see.
5. A pattern is a design that repeats in a regular way.

Exercise 3 — Examples of elements

1. Give two examples of a dot in nature.

2. Give two examples of a line in nature.
3. Give two examples of a shape in nature.
4. What is a geometric shape?
5. **[Review]** What is an organic shape?

Answers

1. Seeds; stars in the sky.
2. Plant stems; the edge of a leaf.
3. Petals; stones.
4. A geometric shape is an exact shape such as a circle, square, or triangle.
5. An organic shape is a free-flowing natural shape such as a cloud or a leaf.

Exercise 4 — Principles of design

1. What are three principles of design?
2. What is balance in design?
3. What is rhythm in design?
4. What is repetition in design?
5. **[Review]** What is a dot?

Answers

1. Balance, rhythm, and repetition.
2. Balance is when the parts of a design feel evenly arranged.
3. Rhythm is a sense of movement created by repeating elements.
4. Repetition is using the same element again and again.
5. A dot is a single small point.

Exercise 5 — Examples of principles

1. Give one example of balance in nature.
2. Give one example of rhythm in manmade design.
3. Give one example of repetition in cloth design.
4. How does repeating dots create rhythm?
5. **[Review]** What is a shape?

Answers

1. The two sides of a face.
2. A row of windows on a building.
3. A repeated motif printed all over a cloth.
4. Repeating dots in a line makes the eye move along the line.
5. A shape is a closed outline you can see.

Exercise 6 — Biomimicry

1. What is biomimicry?
2. The kingfisher beak inspired which machine?
3. Rivers inspired what manmade design?
4. Anthills inspired what manmade design?
5. **[Review]** Name the three elements of design.

Answers

1. Biomimicry is copying ideas from nature to solve human problems.
2. The kingfisher beak inspired the nose of the Shinkansen bullet train.
3. Rivers inspired the design of winding roads.
4. Anthills inspired house designs with natural air flow.
5. Dots, lines, and shapes.

Exercise 7 — Nature and manmade together

1. How is a river similar to a winding road?
2. How is a spider web similar to a fishing net?
3. How are leaf veins similar to a road network?
4. Mention two natural designs and two manmade designs.
5. **[Review]** Name three principles of design.

Answers

1. Both follow the shape of the land as they flow or pass through it.
2. Both use a network of lines to catch things.
3. Both branch out from a main line to reach many points.
4. Natural: spider web, seashell. Manmade: bridge, market stall.
5. Balance, rhythm, and repetition.

Exercise 8 — Scenario and discussion

1. Your school wants a new logo. How does design help?
2. Draw or describe a simple pattern using only dots.
3. Compare a seashell and a spiral staircase.
4. **[Review]** What is biomimicry?
5. **[Review]** What is design?

Answers

1. Design makes the logo clear, memorable, and meaningful.
2. A pattern using only dots might be rows of dots of the same size repeated evenly.
3. Both have a spiral form that grows outward from a centre.
4. Biomimicry is copying ideas from nature to solve human problems.
5. Design is planning how something looks and works.

Summary table

Term	One-line meaning
Design	planning how something looks and works
Elements	dots, lines, shapes
Principles	balance, rhythm, repetition
Biomimicry	copying ideas from nature
Pattern	a design that repeats in a regular way

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